



YILDIZ TECHNICAL UNIVERSITY
MECHATRONICS ENGINEERING
MKT3434
MACHINE LEARNING

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Number: 2106A025



Screenshots

```
D:\3.Sınıf > Makine Öğrenmesi > 2106A025.py > ...
1 import sys
2 import os
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt
6 import seaborn as sns
7
8 # Extensive library imports for machine learning and data processing
9 from PyQt6.QtWidgets import (
10     QApplication, QMainWindow, QWidget, QVBoxLayout, QHBoxLayout,
11     QLabel, QPushButton, QComboBox, QFileDialog,
12     QSpinBox, QDoubleSpinBox, QGroupBox, QScrollArea, QPlainTextEdit,
13     QStatusBar, QProgressBar, QCheckBox, QGridLayout, QMessageBox,
14     QDialog, QLineEdit, QTableWidgetItem, QTableWidgetItem
15 )
16 from PyQt6.QtCore import Qt, QThread, pyqtSignal
17 from PyQt6.QtGui import QFont, QColor
18
19 # Scientific computing and machine learning libraries
20 from sklearn import (
21     datasets, preprocessing, model_selection,
22     metrics, pipeline, feature_selection
23 )
24 from sklearn.linear_model import (
25     LinearRegression, LogisticRegression,
26     Ridge, Lasso, ElasticNet
27 )
28 from sklearn.naive_bayes import (
29     GaussianNB, MultinomialNB, ComplementNB
30 )
31 from sklearn.svm import SVC, SVR
32 from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor
33 from sklearn.ensemble import (
34     RandomForestClassifier, RandomForestRegressor,
35     GradientBoostingClassifier, GradientBoostingRegressor
36 )
37
38 # Advanced application configuration
39 self.setWindowTitle("Advanced Machine Learning Exploration Platform")
40 self.setGeometry(50, 50, 1600, 900)
41 self.setStyleSheet("""
42     QMainWindow { background-color: #f0f0f0; }
43     QGroupBox {
44         border: 2px solid #000000;
45         border-radius: 5px;
46         margin-top: 10px;
47         font-weight: bold;
48     }
49     QGroupBox::title {
50         subcontrol-origin: margin;
51         left: 16px;
52         padding: 0 3px;
53     }
54 """)
55
56 # Core data management attributes
57 self.dataset_manager = {
58     'training_inputs': None,
59     'testing_inputs': None,
60     'training_targets': None,
61     'testing_targets': None,
62     'feature_names': [],
63     'target_name': ''
64 }
65
66 # Model tracking and configuration
67 self.model_registry = {
68     'current_model': None,
69     'model_history': []
70 }
71
72 # CategoryInfo : ObjectNotFound: (C:/Users/CLAL A...n310/python.exe:String) [], CommandNotFoundException
73 # FullyQualifiedErrorId : CommandNotFoundException
74
75 PS C:\Users\CLAL AKYOL>
```

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102 PS C:\Users\CLAL AKYOL> "C:/Users/CLAL AKYOL/AppData/Local/Programs/Python/Python310/python.exe" "d:/3.Sınıf/Makine Öğrenmesi/2106A025.py"
103 & : The term 'C:/Users/CLAL AKYOL/AppData/Local/Programs/Python/Python310/python.exe' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path is used, verify that the path is correct and try again.
104 At line:1 char:3
105 + & "C:/Users/CLAL AKYOL/AppData/Local/Programs/Python/Python310/python ...
106 ~~~~~
107 + CategoryInfo          : ObjectNotFound: (C:/Users/CLAL A...n310/python.exe:String) [], CommandNotFoundException
108 + FullyQualifiedErrorId : CommandNotFoundException
109
110 PS C:\Users\CLAL AKYOL>
```



NOT: Visual Studio Uygulamasından aldığım hata nedeniyle kodla yaptığım denemeleri maalesef görselleştiremedim. Dersin teorik anlamda gerekliliklerini anlasam da koda dökemediğimden emin olamayarak ödevi paylaşıyorum. Bir sonraki ödevde tüm sorunları çözerek düzgün denemeleri ve çalıştırmaları yaptığım bir teslim gerçekleştireceğim.